

SDS 711: PROGRAMMING FOR DATA SCIENCE

Fall Semester 2026

COURSE DESCRIPTION: This course introduces students to the fundamental principles and practices of computer programming for data science applications. It covers the complete data science workflow, from data acquisition and databases to data wrangling, visualization, machine learning, and software development. Students will gain practical skills in modern tools such as SQL, Python, R, and Git, while learning to write efficient code, analyse data, and communicate insights effectively. The course assumes no prior programming knowledge and is designed to build computational thinking and problem-solving skills for real-world data-driven decision-making.

LECTURER: Bubacarr Bah, bubacarr.bah1@lshtm.ac.uk

CLASS REP.: Musa Kebbeh, kebbehmusa71@gmail.com

LECTURE TIMINGS AND VENUE: Saturdays: 8:30am – 11:30am GMT, LR 5, School of ICT, Faraba.

TUTORIAL SESSIONS TIMINGS AND VENUE: Would be arranged with the class, if needed.

COURSE INFORMATION: This course will use **GitHub**, **Google Drive**, and **WhatsApp**. The syllabus, class announcements, homeworks, solutions and supplementary learning materials will be posted these channels; while homework submissions will be strictly via **GitHub**.

COURSE TEXTBOOK:

1. Primary text
 - (a) *Data Science: A First Introduction* (free online)
2. Supplementary texts:
 - (a) *Python for Data Analysis* (Python and data wrangling)
 - (b) *Getting Started with SQL* (Databases)
 - (c) *Introduction to Machine Learning with Python* (Machine Learning)

ASSESSMENT: The assessment for this course will comprise of quizzes or homeworks, two midterm exams, and a cumulative final exam. The final score will be computed according to the following breakdown:

1. Continuous Assessment (50% of total mark):
 - (a) Assignments / Coding Tasks – 15 %
 - (b) Tests and Quizzes – 10 %
 - (c) Practical Laboratory Exercises – 15%
 - (d) Class Participation / Tutorials / Presentations – 10%
2. Final Examination (50% of total mark):
 - (a) Formal Written Examination (theory and applied programming problems) – 50%

The continuous assessment and final examination scores will be combined for a total of 100%. Your final score will be rounded to the nearest integer and you will be assigned a final letter grade according to the standard UTG grading system.

ASSIGNMENTS / CODING TASKS: There will be two (2) homework assignments, which would involve some coding. Each assignment will be shared with the class approximately two week before it is due. The second assignment will be a group mini project, see below. Assignments will be submitted online via GitHub by the due date agreed in class; see schedule below for a guide. There will be no make-up homework, so a

missed homework counts as zero. No homework assignment score will be dropped. The two scores will be averaged to determine the 15% assignments / coding tasks score.

MINI PROJECT: We will have one (1) mini project assignment (HW 2) towards the end of the semester, see schedule below. This will be a group project. The project will be computational and students would be required to hand in code and make a presentation.

TESTS AND QUIZZES: There will be one (1) test and two (2) quizzes. The quizzes will be typically held in the first 15 minutes of the class, while the test will take a little longer.

- The lowest quiz scores out of the two quizzes will be dropped to allow for valid excuses.
- The remaining score will be combined with the test to determine the 10% tests and quizzes score.
- There will be no make-up quizzes except for religious holidays. So a missed quiz counts as zero, hence qualifies as one of your dropped scores.

PRACTICAL LABORATORY EXERCISES: Two (2) practical exercises will be held in class, see schedule below. This will typically be coding exercises. Similar to the quizzes the best out of the 2 scores will be taken to represent your score for this category.

CLASS PARTICIPATION / TUTORIALS / PRESENTATIONS: We will monitor classroom participation and performance during (group) presentations to award the 10% score of this category.

FINAL EXAM: There will be a cumulative final exam that will count for 50% of your overall score. The date and location of the exam will be determined by the Office of the Registrar.

EMAIL GUIDELINES: In order to make it easier to provide prompt responses to your emails, when emailing me, please include the class name “SDS 711 ” in the subject line and sign the e-mail with your full name. If you need to make an appointment with me outside of my regular office hours, please suggest several time slots when you will like to meet.

COURSE OUTLINE: Details of the sections outlined in the **course schedule**.

1. Introduction to programming concepts and computational thinking
2. Version control with Git and GitHub
3. Python programming fundamentals
4. R programming fundamentals
5. Data types, control structures, and functions
6. Working with files and data input/output
7. Databases and SQL for data science
8. Data wrangling and preprocessing
9. Exploratory data analysis
10. Data visualization techniques
11. Introduction to machine learning
12. Model evaluation and validation
13. Reproducible workflows in data science
14. Communicating results to stakeholders

COURSE SCHEDULE: We would roughly work through the course outline according to the schedule in the table below. The numbers indicate sections of the course outline (as soon in the next page). This a tentative schedule and we may alter it if need be to improve learning experience.

Date	Section	Assessment	Date	Section	Assessment
Sep 19	1		Nov 21	11	<i>HW 2 – mini project</i>
Sep 26	2	<i>Ex 1</i>	Nov 28	12	
Oct 03	3		Dec 05	13, 14	<i>Quiz 2, Last lecture</i>
Oct 10	4		Dec 12		<i>Presentations</i>
Oct 17	5, 6	<i>Quiz 1</i>	Dec 19		<i>Final Exam</i>
Oct 24	7		Dec 26		
Oct 31	8	<i>Test 1</i>	Jan 02		
Nov 07	9	<i>HW 1</i>	Jan 09		
Nov 14	10	<i>Ex 2</i>	Jan 16		